Table 1: **Reasoning post-training.** Every row receives the same OpenThoughts3-30K update. Benchmark columns and the suite average are the endpoint, and Δ is the change in the suite average from each row's starting checkpoint. Evaluation uses Protocol B rather than Table **??**'s Protocol A. Scores are on the full benchmarks, which include items identified in the training data (Section **??**), so they are not uniformly held-out estimates. Starting scores and SDs are in Appendix Table **??**, and the interval method in Appendix **??**.

	Benchmark means after OT3-30K											**Suite average**	
Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg	Δ [95% CI]
(a) Main comparison: Tulu3 SFT → OT3-30K													
300B-token models													
MV	11.1	16.7	46.7	71.0	49.0	51.0	11.5	28.5	28.8	18.8	42.1	34.1	+10.7 [+8.2, +13.2]
FineWeb-Edu	0.0	0.0	1.7	0.6	6.2	0.0	0.0	0.0	23.9	1.9	33.4	6.2	−1.3 [−2.3, −0.5]
Nemotron-CC-HQ	0.0	0.0	0.0	0.3	2.9	0.2	0.0	0.0	25.1	1.1	35.5	5.9	−2.1 [−3.1, −1.2]
Comma0.1	0.0	0.0	0.0	1.3	13.4	1.4	0.0	0.1	26.8	2.6	36.8	7.5	−4.0 [−5.1, −3.0]
External references, uncontrolled													
SmolLM2	0.0	0.0	1.7	6.6	23.1	14.2	0.2	10.5	26.3	3.6	41.7	11.6	−3.8 [−5.1, −2.4]
Qwen2.5	4.4	1.1	26.7	44.9	57.6	21.7	2.8	12.3	25.9	14.1	40.6	22.9	−0.7 [−2.6, +1.3]
Qwen3	0.0	0.0	1.7	10.9	25.7	8.5	0.5	6.7	29.0	2.0	38.8	11.3	−17.2 [−18.7, −15.7]
(b) MV starting-point controls													
16K extended	8.9	25.6	42.5	69.8	58.8	54.7	13.4	28.8	30.1	18.4	40.9	35.6	+7.6 [+5.2, +10.0]
Tulu3 SFT + DPO	0.0	4.4	17.5	34.4	44.4	46.1	8.6	34.5	27.1	11.5	49.2	25.3	+0.5 [−1.5, +2.7]

Table 2: **Phase 2 changes on the suite average, with intervals.** The change of every run between the checkpoints of Table **??**, each against the checkpoint it was trained from: the updates of Table 1 and MixtureVitae's two Tulu3 steps. Brackets are 95% paired bootstrap intervals over the questions, each question scored as its mean over the runs. They do not include differences between training runs. The code tasks have no per-question verdicts in the records and enter with the conservative approximation of Appendix **??**. **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction over the 11 changes. The last column uses runs 1 and 2 only, leaving out run 0, which repeats an earlier generation in some cells.

Run	**Change, 95% interval**	q	**Runs 1 and 2 only**
MixtureVitae-16K			
→ OT30K (without Tulu3)	**+7.6** [+5.2, +10.0]	< 0.001	+5.7 [+3.3, +8.0]
→ SFT	*−4.5* [−6.6, −2.5]	< 0.001	−6.1 [−8.3, −3.8]
→ SFT → OT30K	**+10.7** [+8.2, +13.2]	< 0.001	+10.7 [+8.1, +13.4]
→ SFT → DPO	+1.3 [−0.5, +3.2]	0.194	+0.9 [−1.0, +2.7]
→ SFT → DPO → OT30K	+0.5 [−1.5, +2.7]	0.644	+0.8 [−1.2, +3.0]
FineWeb-Edu-16K			
→ OT30K	*−1.3* [−2.3, −0.5]	0.005	−1.2 [−2.1, −0.3]
Nemotron-CC-HQ-16K			
→ OT30K	*−2.1* [−3.1, −1.2]	< 0.001	−2.2 [−3.2, −1.3]
Comma0.1-16K			
→ OT30K	*−4.0* [−5.1, −3.0]	< 0.001	−3.9 [−5.0, −2.8]
SmolLM2-16K			
→ OT30K	*−3.8* [−5.1, −2.4]	< 0.001	−3.4 [−4.8, −2.1]
Qwen2.5-1.5B			
→ OT30K	−0.7 [−2.6, +1.3]	0.536	−0.6 [−2.7, +1.6]
Qwen3-1.7B			
→ OT30K	*−17.2* [−18.7, −15.7]	< 0.001	−17.4 [−19.2, −15.8]

Table 3: **The OT3 exclusion: the two gains and their difference.** The change of each model of Table **??** from its own 16K extended checkpoint under the identical OT3-30K update, and the first change minus the second, which is how much more the model with OpenThoughts3 in pretraining gains. The two bases differ before the update, so the difference of the gains is not the difference of the two endpoints. Brackets are 95% paired bootstrap intervals over the questions, each question scored as its mean over the runs. They do not include differences between training runs. The code tasks have no per-question verdicts in the records and enter with the conservative approximation of Appendix **??**. **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction over the six results of the first two columns. The last two columns use runs 1 and 2 only, leaving out run 0, which repeats an earlier generation in some cells.

	All three runs		**Runs 1 and 2 only**	
	Suite average	**MATH500**	**Suite average**	**MATH500**
OT3 in pretraining, gain	**+7.6** [+5.2, +10.0]	**+23.3** [+20.3, +26.0]	+5.7 [+3.3, +8.0]	+17.0 [+13.7, +20.4]
OT3 removed from pretraining, gain	+1.7 [−0.2, +3.6]	**+8.7** [+6.0, +11.6]	+1.6 [−0.4, +3.6]	+11.4 [+8.2, +14.8]
Difference of the two gains	**+5.9** [+2.2, +9.9]	**+14.5** [+10.9, +18.4]	+4.0 [+0.1, +8.0]	+5.6 [+1.4, +10.3]

Table 4: **Phase 2 changes by domain.** Every run of Tables **??**, **??**, **??** and **??**, against the checkpoint it was trained from, on the mean of each domain's tasks (Table **??** gives each task's domain) and of every task but mathematics, in three panels. Brackets are 95% intervals (Appendix **??**). **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction within each column, over all three panels. A dash marks a domain with a task that the run or its parent was not evaluated on.

(a) Every base from its Tulu3 SFT checkpoint

From	Update	Math	Code	Sci. and QA	IFEval	All but math
MixtureVitae-16K						
Tulu3 SFT	OpenThoughts3	**+21.9** [+18.0, +26.2]	+2.4 [−3.8, +8.6]	**+6.2** [+2.2, +10.0]	*−11.9* [−16.1, −8.1]	+1.3 [−2.2, +4.5]
Tulu3 SFT	AceReason	**+22.0** [+18.1, +26.0]	+7.1 [+0.6, +13.0]	**+5.9** [+1.9, +9.8]	*−12.4* [−16.3, −8.4]	+3.4 [+0.0, +6.8]
Tulu3 SFT	Nemotron-Cascade	**+25.4** [+21.5, +29.7]	**+7.4** [+1.3, +13.5]	**+6.9** [+3.0, +10.7]	*−9.9* [−13.6, −6.2]	**+4.4** [+0.8, +7.6]
Tulu3 SFT	Superior-Reasoning	**+13.5** [+10.9, +16.7]	*−8.2* [−13.9, −2.5]	+0.1 [−4.0, +4.3]	*−12.8* [−16.9, −8.5]	*−6.2* [−9.3, −2.9]
FineWeb-Edu-16K						
Tulu3 SFT	OpenThoughts3	−0.1 [−0.8, +0.5]	*−1.1* [−1.9, −0.3]	−3.0 [−7.2, +1.2]	*−4.7* [−8.1, −1.3]	*−2.3* [−4.0, −0.9]
Tulu3 SFT	AceReason	−0.1 [−0.7, +0.3]	−0.6 [−2.0, +0.9]	−0.5 [−3.8, +2.8]	*−5.2* [−8.5, −2.1]	−1.3 [−2.9, +0.1]
Tulu3 SFT	Nemotron-Cascade	+0.1 [−0.7, +0.9]	−0.8 [−2.1, +0.4]	+0.5 [−3.4, +4.3]	*−4.0* [−7.2, −0.9]	−0.9 [−2.5, +0.6]
Tulu3 SFT	Superior-Reasoning	−0.5 [−1.1, +0.0]	*−1.1* [−1.9, −0.2]	−2.1 [−5.7, +1.7]	+0.9 [−2.3, +4.5]	−1.1 [−2.5, +0.3]
Nemotron-CC-HQ-16K						
Tulu3 SFT	OpenThoughts3	*−1.8* [−2.9, −1.0]	*−1.7* [−2.9, −0.6]	−2.8 [−6.3, +1.1]	*−3.7* [−7.2, −0.2]	*−2.4* [−3.9, −0.9]
Tulu3 SFT	AceReason	−0.7 [−1.8, +0.1]	−1.2 [−2.8, +0.5]	−0.5 [−4.0, +3.0]	*−5.2* [−8.5, −1.8]	−1.6 [−3.2, −0.1]
Tulu3 SFT	Nemotron-Cascade	−1.0 [−2.0, +0.0]	−1.2 [−2.8, +0.6]	−1.4 [−4.9, +1.9]	*−5.6* [−9.1, −2.4]	*−2.0* [−3.5, −0.4]
Tulu3 SFT	Superior-Reasoning	*−1.5* [−2.6, −0.7]	*−1.7* [−3.0, −0.3]	−2.3 [−5.7, +0.8]	+0.1 [−3.4, +3.7]	−1.6 [−3.0, −0.2]
Comma0.1-16K						
Tulu3 SFT	OpenThoughts3	−0.3 [−1.2, +0.4]	*−9.9* [−12.6, −7.2]	−1.3 [−4.4, +1.9]	*−10.1* [−13.5, −6.6]	*−7.0* [−8.8, −5.2]
Tulu3 SFT	AceReason	−0.4 [−1.2, +0.4]	*−6.7* [−10.3, −3.2]	−0.6 [−3.4, +2.2]	*−11.5* [−14.9, −8.0]	*−5.5* [−7.6, −3.5]
Tulu3 SFT	Nemotron-Cascade	+0.0 [−0.9, +0.8]	*−5.4* [−9.2, −1.8]	+0.9 [−2.1, +3.8]	*−7.8* [−11.1, −4.4]	*−3.7* [−5.8, −1.4]
Tulu3 SFT	Superior-Reasoning	−0.5 [−1.3, +0.2]	*−9.1* [−12.0, −6.2]	−2.0 [−4.8, +0.9]	*−6.2* [−9.9, −2.8]	*−6.3* [−8.1, −4.5]
SmolLM2-16K						
Tulu3 SFT	OpenThoughts3	−0.6 [−1.3, +0.1]	*−7.6* [−12.1, −3.0]	−1.7 [−3.3, +0.0]	*−12.2* [−14.9, −9.6]	*−6.4* [−8.8, −4.0]
Tulu3 SFT	AceReason	–	−2.1 [−6.9, +2.8]	+1.1 [−1.0, +3.3]	*−7.2* [−9.7, −4.6]	−1.9 [−4.6, +0.8]
Tulu3 SFT	Nemotron-Cascade	**+3.5** [+2.4, +4.5]	+1.9 [−3.2, +6.8]	−0.4 [−3.6, +2.5]	*−11.4* [−14.7, −8.3]	−1.1 [−3.9, +1.7]
Tulu3 SFT	Superior-Reasoning	**+3.1** [+2.2, +3.9]	*−8.7* [−13.0, −4.2]	−0.3 [−2.3, +1.7]	*−12.5* [−15.8, −9.4]	*−6.5* [−9.0, −4.1]
Qwen2.5-1.5B						
Tulu3 SFT	OpenThoughts3	**+10.4** [+8.1, +13.0]	*−17.6* [−22.8, −12.1]	**+3.6** [+1.1, +6.1]	*−14.3* [−17.8, −10.7]	*−9.9* [−12.8, −7.1]
Qwen3-1.7B						
Tulu3 SFT	OpenThoughts3	*−13.0* [−14.5, −11.5]	*−31.3* [−35.6, −26.9]	*−5.1* [−7.8, −2.3]	*−20.3* [−23.6, −17.1]	*−20.7* [−23.2, −18.1]

Table 4: (continued) Panels (b) and (c).

(b) MixtureVitae from its 16K extended checkpoint and its DPO checkpoint

From	Update	Math	Code	Sci. and QA	IFEval	All but math
MixtureVitae-16K						
16K extended checkpoint	OpenThoughts3	**+13.2** [+9.8, +16.7]	+4.3 [−1.7, +10.6]	**+5.4** [+1.4, +9.3]	*−5.7* [−8.8, −2.9]	+3.0 [−0.4, +6.6]
16K extended checkpoint	AceReason	**+10.6** [+8.2, +13.1]	**+8.5** [+2.4, +14.7]	**+6.4** [+2.6, +10.2]	*−6.4* [−9.3, −3.6]	**+5.3** [+1.9, +8.7]
16K extended checkpoint	Nemotron-Cascade	**+11.4** [+8.7, +14.3]	**+7.0** [+1.2, +13.2]	**+5.8** [+1.9, +9.8]	*−3.5* [−6.2, −0.7]	**+4.9** [+1.4, +8.2]
16K extended checkpoint	Superior-Reasoning	**+3.7** [+1.1, +6.1]	*−7.8* [−13.7, −1.7]	−0.6 [−4.9, +3.7]	*−5.0* [−7.9, −2.2]	*−4.9* [−8.2, −1.5]
Tulu3 SFT + DPO	OpenThoughts3	+1.4 [−0.8, +3.8]	+0.3 [−5.7, +6.4]	+1.6 [−0.5, +3.6]	*−5.3* [−7.9, −2.9]	−0.2 [−3.3, +3.0]
Tulu3 SFT + DPO	AceReason	**+20.0** [+16.1, +24.3]	+5.3 [−0.9, +11.3]	**+7.1** [+3.2, +11.0]	*−12.5* [−16.1, −9.0]	+2.9 [−0.5, +6.3]
Tulu3 SFT + DPO	Nemotron-Cascade	**+23.3** [+19.6, +27.3]	+5.5 [−0.5, +11.8]	**+7.7** [+3.7, +11.4]	*−10.7* [−14.1, −7.5]	+3.5 [+0.1, +6.8]
Tulu3 SFT + DPO	Superior-Reasoning	**+11.9** [+9.4, +14.8]	*−9.9* [−15.5, −4.2]	−0.4 [−4.4, +3.6]	*−13.4* [−17.4, −9.6]	*−7.3* [−10.6, −3.9]

(c) Controls: the Tulu3 steps, and MixtureVitae without OpenThoughts3

From	Update	Math	Code	Sci. and QA	IFEval	All but math
MixtureVitae-16K						
16K extended checkpoint	Tulu3 SFT	*−10.9* [−13.7, −8.2]	−0.1 [−6.1, +6.1]	−1.2 [−3.4, +1.0]	**+7.4** [+4.3, +10.6]	+0.8 [−2.2, +3.9]
Tulu3 SFT	Tulu3 DPO	**+1.8** [+0.2, +3.4]	+1.6 [−4.4, +7.4]	+0.1 [−0.7, +0.9]	+0.6 [−1.9, +3.2]	+0.9 [−2.2, +3.9]
MixtureVitae without OT3-16K						
16K extended checkpoint	OpenThoughts3	**+3.5** [+1.8, +5.6]	+1.0 [−5.2, +6.9]	+0.6 [−1.1, +2.3]	*−3.3* [−5.3, −1.2]	+0.2 [−2.9, +3.2]

Table 5: **MixtureVitae against each reference before and after the reasoning update.** MV minus the reference, in points, on the macro average of the five mathematics tasks (AIME24, AIME25, AMC23, MATH500, GSM8K), on the macro average of the three contest tasks (AIME24, AIME25, AMC23, 100 problems), and on IFEval, with both models at their Tulu3 SFT checkpoints and after OT30K from them. The widening is MV's gain under OT30K minus the reference's, so it says whether the gap after the update is larger than before it. Protocol B, as in Table 1. Brackets are 95% paired bootstrap intervals over the questions, each question scored as its mean over the runs. They do not include differences between training runs. **Bold** marks a significant difference in MV's favour and *italic* one in the reference's favour after Benjamini–Hochberg correction within each of mathematics, contest mathematics and IFEval.

	Mathematics			Contest mathematics		IFEval
Reference	Tulu3 SFT	After OT30K	Widening	Tulu3 SFT	After OT30K	After OT30K
FineWeb-Edu	**+15.1** [+13.7, +16.6]	**+37.2** [+33.3, +41.5]	**+22.0** [+18.2, +26.3]	**+2.8** [+0.6, +5.0]	**+24.3** [+17.5, +31.1]	**+8.7** [+7.0, +10.4]
Nemotron-CC-HQ	**+14.5** [+12.8, +16.2]	**+38.2** [+34.3, +42.3]	**+23.7** [+20.0, +27.8]	+1.5 [−1.1, +4.0]	**+24.8** [+18.4, +31.5]	**+6.6** [+5.0, +8.3]
Comma0.1	**+13.7** [+12.2, +15.4]	**+35.9** [+31.9, +40.2]	**+22.2** [+18.5, +26.3]	+1.9 [−0.3, +4.3]	**+24.8** [+18.1, +31.7]	**+5.3** [+3.8, +6.9]
SmolLM2	**+10.1** [+8.8, +11.6]	**+32.6** [+28.6, +36.8]	**+22.5** [+18.6, +26.7]	**+2.6** [+0.6, +4.8]	**+24.3** [+17.3, +31.3]	+0.4 [−2.7, +3.6]
Qwen2.5	+0.4 [−1.1, +2.0]	**+11.9** [+8.7, +15.6]	**+11.5** [+7.6, +15.7]	+0.9 [−1.4, +3.1]	**+14.1** [+8.6, +19.8]	+1.5 [−0.3, +3.4]
Qwen3	*−3.6* [−5.2, −2.0]	**+31.2** [+27.3, +35.4]	**+34.9** [+30.5, +39.2]	−1.4 [−3.8, +1.1]	**+24.3** [+18.3, +31.0]	**+3.3** [+0.7, +5.9]

Table 6: **The mathematics gap after OT30K with each task left out.** MV minus the reference after OT30K on the macro average of the mathematics tasks of Table 5, with the task named in the column left out. Brackets are 95% paired bootstrap intervals over the questions, each question scored as its mean over the runs. They do not include differences between training runs. The intervals are robustness diagnostics, chosen after seeing every removal, and are not corrected for multiple comparisons.

Reference	Without A24	Without A25	Without AMC	Without M500	Without GSM8K
FineWeb-Edu	+43.7 [+39.1, +48.3]	+42.3 [+38.2, +46.6]	+35.2 [+31.8, +39.6]	+28.9 [+23.7, +34.2]	+35.8 [+30.6, +41.0]
Nemotron-CC-HQ	+45.0 [+40.5, +49.7]	+43.6 [+39.6, +47.7]	+36.1 [+32.4, +40.6]	+30.1 [+25.2, +35.4]	+36.3 [+31.3, +41.5]
Comma0.1	+42.2 [+37.8, +46.8]	+40.8 [+36.8, +44.9]	+33.3 [+29.6, +37.6]	+27.5 [+22.8, +32.6]	+36.0 [+31.1, +41.4]
SmolLM2	+38.0 [+33.2, +42.8]	+36.6 [+32.5, +41.0]	+29.5 [+26.0, +33.8]	+24.7 [+19.8, +29.9]	+34.3 [+29.3, +39.7]
Qwen2.5	+13.3 [+9.3, +17.4]	+11.0 [+7.7, +14.7]	+9.9 [+6.4, +13.6]	+8.4 [+4.2, +12.9]	+17.1 [+12.9, +21.6]
Qwen3	+36.3 [+31.6, +40.8]	+34.9 [+31.0, +39.1]	+27.8 [+24.1, +31.7]	+24.0 [+19.2, +29.1]	+33.2 [+28.1, +38.4]

Table 7: **Answers near the token limit.** Share of the answers, in percent over the three runs, whose length, counted with each checkpoint's own tokenizer, is at least 95% of the generation limit of Table **??**, since re-encoding a sampled answer can give a few percent fewer tokens than were generated. This is a length criterion, not the reason a generation stopped. The rows are those of Tables **??** and **??**.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE
MixtureVitae-16K											
16K extended checkpoint	21.1	13.3	14.2	6.1	0.0	0.0	9.8	0.3	0.0	0.1	1.1
→ OT30K (without Tulu3)	77.8	57.8	40.0	18.1	0.3	38.2	80.6	42.9	32.2	46.9	21.5
→ SFT	0.0	10.0	1.7	0.3	0.5	0.9	0.2	0.2	0.0	0.5	3.7
→ SFT → OT30K	70.0	65.6	36.7	15.9	3.8	22.8	80.6	42.2	31.3	47.0	17.4
→ SFT → DPO	0.0	7.8	0.0	0.3	0.5	0.2	0.4	0.1	0.0	0.5	3.1
→ SFT → DPO → OT30K	24.4	31.1	9.2	12.5	1.3	6.9	4.4	1.7	15.7	12.2	24.0
FineWeb-Edu-16K											
Tulu3 SFT	3.3	12.2	0.0	2.2	1.4	2.4	5.3	1.8	0.2	3.2	1.9
→ OT30K	86.7	91.1	84.2	82.0	14.5	97.4	99.6	96.7	84.7	83.1	79.9
Nemotron-CC-HQ-16K											
Tulu3 SFT	0.0	0.0	0.0	0.0	4.5	1.2	4.4	2.3	0.0	0.0	5.4
→ OT30K	100.0	100.0	99.2	97.2	36.7	99.2	99.4	99.5	94.8	96.4	89.5
Comma0.1-16K											
Tulu3 SFT	3.3	11.1	3.3	1.8	2.0	0.0	0.9	0.5	0.0	1.7	3.2
→ OT30K	85.6	100.0	86.7	79.0	11.9	91.1	99.3	95.1	68.0	79.0	71.9
SmolLM2-16K											
Tulu3 SFT	1.1	7.8	0.0	0.8	3.8	0.6	0.5	0.3	0.0	1.5	3.5
→ OT30K	77.8	91.1	72.5	61.1	25.2	27.8	61.7	47.5	3.4	74.2	6.9
Qwen2.5-1.5B											
Tulu3 SFT	2.2	1.1	1.7	0.2	0.3	0.0	0.4	0.1	0.0	0.5	3.6
→ OT30K	64.4	73.3	48.3	57.1	8.6	90.2	93.6	84.0	63.3	42.6	91.3
Qwen3-1.7B											
Tulu3 SFT	1.1	1.1	0.0	0.4	0.3	0.2	0.3	0.0	0.0	0.3	3.3
→ OT30K	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
MixtureVitae without OT3-16K											
Base without OT3, before reasoning post-training	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.3
OT3 removed from pretraining, → OT30K	0.0	0.0	0.0	0.0	0.4	0.0	26.1	0.2	0.0	0.0	2.5

Table 8: **Answers with no final answer, and code answers with no code block.** Share of the answers, in percent over the three runs. For the mathematics and science tasks, the share from which the grader extracts no final answer to compare with the reference, so that the answer is scored wrong. For the code tasks, the share with no code block. The checker then runs the whole answer, so such an answer is not scored wrong for that reason alone. GSM8K and GPQA-Diamond are left out, because their graders take an answer from every response, and IFEval has no answer to extract. A dash marks a cell in which the records do not hold the extracted answer or code. The rows are those of Tables **??** and **??**.

Model	**A24**	**A25**	**AMC**	**M500**	**HE**	**LCB**	**MBPP**	**JEE**
MixtureVitae-16K								
16K extended checkpoint	18.9	16.7	15.0	6.9	0.0	5.8	0.0	3.9
→ OT30K (without Tulu3)	71.1	52.2	35.0	15.2	4.3	32.6	5.4	39.7
→ SFT	24.4	16.7	19.2	15.3	–	0.0	–	23.0
→ SFT → OT30K	65.6	58.9	30.8	13.0	5.3	33.6	7.0	38.9
→ SFT → DPO	23.3	21.1	11.7	13.8	0.0	0.0	0.0	20.6
→ SFT → DPO → OT30K	16.7	15.6	4.2	5.8	0.0	0.9	0.1	4.7
FineWeb-Edu-16K								
Tulu3 SFT	58.9	63.3	55.0	58.2	0.4	0.7	0.9	43.9
→ OT30K	77.8	90.0	70.8	71.3	16.1	60.5	13.2	71.0
Nemotron-CC-HQ-16K								
Tulu3 SFT	36.7	32.2	31.7	46.7	0.0	32.4	0.2	35.4
→ OT30K	93.3	92.2	94.2	82.9	4.1	55.0	5.8	79.2
Comma0.1-16K								
Tulu3 SFT	47.8	23.3	50.0	39.6	1.2	0.1	0.0	15.8
→ OT30K	80.0	97.8	78.3	67.1	9.1	53.3	5.0	66.9
SmolLM2-16K								
Tulu3 SFT	31.1	23.3	22.5	22.8	0.2	0.0	0.0	10.6
→ OT30K	81.1	94.4	65.8	54.1	0.8	6.4	3.4	59.8
Qwen2.5-1.5B								
Tulu3 SFT	–	–	–	11.8	0.0	0.0	0.1	27.1
→ OT30K	42.2	55.6	20.8	21.3	3.0	23.6	2.3	19.3
Qwen3-1.7B								
Tulu3 SFT	5.6	1.1	12.5	16.2	0.0	0.0	0.0	10.5
→ OT30K	95.6	90.0	85.0	76.2	0.8	36.7	7.1	79.4
MixtureVitae without OT3-16K								
Base without OT3, before reasoning post-training	–	–	–	6.1	0.0	0.0	0.0	4.5
OT3 removed from pretraining, → OT30K	13.3	8.9	3.3	6.5	0.0	12.3	0.1	19.4